



M.TECH FINAL THESIS DEFENSE

Deadline-Based Quantum-Inspired Genetic Algorithm for Scheduling Workflows

Optimizing Resource Allotments in Cloud-Fog Environments

TECHNO MAIN SALT LAKE • DEPT. OF COMPUTER SCIENCE & ENGINEERING

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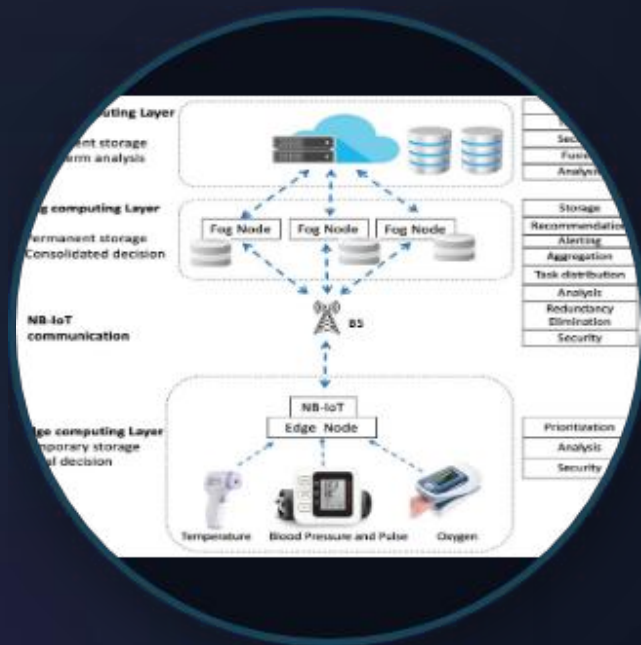
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Introduction to Cloud-Fog Computing



Bridging the Edge and Cloud

The explosion of IoT devices requires low-latency, bandwidth-efficient architectures. Combining edge proximity with massive cloud resources creates a critical scheduling hierarchy.

-  **Fog Layer:** Located near edge devices. Highly localized, offering minimal transmission delay, but strictly constrained in computing capacity.
-  **Cloud Layer:** Centralized and virtually limitless resources, but offloading computationally intensive tasks suffers from severe network congestion.
-  **Workflow Challenge:** Scientific processing consists of interdependent tasks requiring optimized allocations to minimize transmission costs.

| Challenges in Workflow Scheduling

Efficient task orchestration across hybrid infrastructures involves managing conflicting requirements under strict performance guarantees.



Complexity

*Scheduling complex workflows on heterogeneous computing nodes is classified as an **NP-Complete** optimization problem, requiring robust meta-heuristics.*



DAG Dependencies

Tasks are modeled as a Directed Acyclic Graph (DAG) with strict parent-child constraints, meaning communication delay dominates total execution time.



Local Optima Traps

Classical scheduling models (e.g., standard heuristics) suffer from slow convergence speeds and easily get trapped in sub-optimal local regions.

Mathematical Formulation & Objectives

Objective & Constraints

The primary focus is minimizing the overall execution cost (tec_w) while ensuring the completed schedule (m_w) meets the strict user-defined deadline constraint (D_w):

$$\text{Minimize: } tec_w \text{ subject to } m_w < D_w$$

The workflow execution cost comprises both node computation tariffs ($\$cmp_w$) and inter-resource transmission costs ($\$ccp_w$).

Cost Components Model

The mathematical representation of total execution costs matches the following criteria:

$$tec_w = cmp_w + ccp_w$$

- Computation cost relies on resource processing unit price multiplied by the elapsed execution duration.
- Communication overhead applies if dependent task pairs are scheduled on different physical node locations.

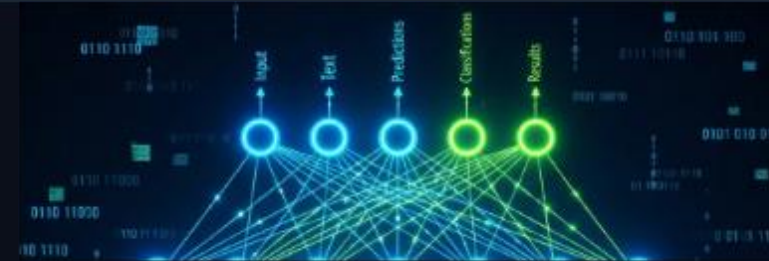
The Baseline: Quantum Genetic Algorithm

Superposition & State Collapse

Classical QGA uses Quantum bits (Q-bits) in a state of superposition to probabilisticly explore the extensive workflow allocation search space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{ where } \alpha^2 + \beta^2 = 1$$

- ⚠ **High Variance:** Single-shot measurements trigger stochastic noise during collapse, leading to unstable allocations.
- 🔗 **Premature Convergence:** The lack of a directional update gate causes classical QGA to stagnate on complex DAG setups.



DQGA Execution Flow



Innovation 1: Cumulative Based Measure Function

Replacing single-shot quantum collapse with statistical stability filters to guarantee highly reproducible allocations.



Probabilistic Integration

Converts quantum amplitudes (alpha, beta) into definitive state probabilities, generating a cumulative frequency vector (cf_k) across basis states.



Multi-Iteration Sampling

*Performs **15** internal statistical sampling rounds per task. This absorbs minor stochastic noise, preventing early random divergent collapses.*



Majority Voting

*Identifies the final physical node assignment via statistical mode (majority winner). Achieves a **12%** reduction in makespan result variance.*

Innovation 2: Task-Aware Clustered Crossover

Standard Crossover Issues

The Problem of Broken IDs:

Bit-wise crossovers randomly split a single task's representation, causing mutated, invalid node IDs or impossible schedules.

✖ Standard crossover creates high selection pressure, degrading the quality of the chromosome population.

QACCR Crossover Integration

Our Multi-Level Solution:

Crossover boundaries are restricted to multiples of the Bits Per Node (BPN). Full task blocks are moved as a single cohesive unit.

✔ Preserves level-by-level task cluster execution context without creating corrupted binary offsets.

Adaptive Quantum Rotation Gates

Dynamic Update Mechanism

Unlike the destructive nature of classical genetic mutations, the DQGA population evolves smoothly by shifting the qubit phases toward the current global best using a rotation matrix:

$$U(\Delta\theta) = \begin{bmatrix} \cos(\Delta\theta) & -\sin(\Delta\theta) \\ \sin(\Delta\theta) & \cos(\Delta\theta) \end{bmatrix}$$

Guided Mathematical Search

The selection of the rotation step (Delta, theta) relies on comparative cost metrics:

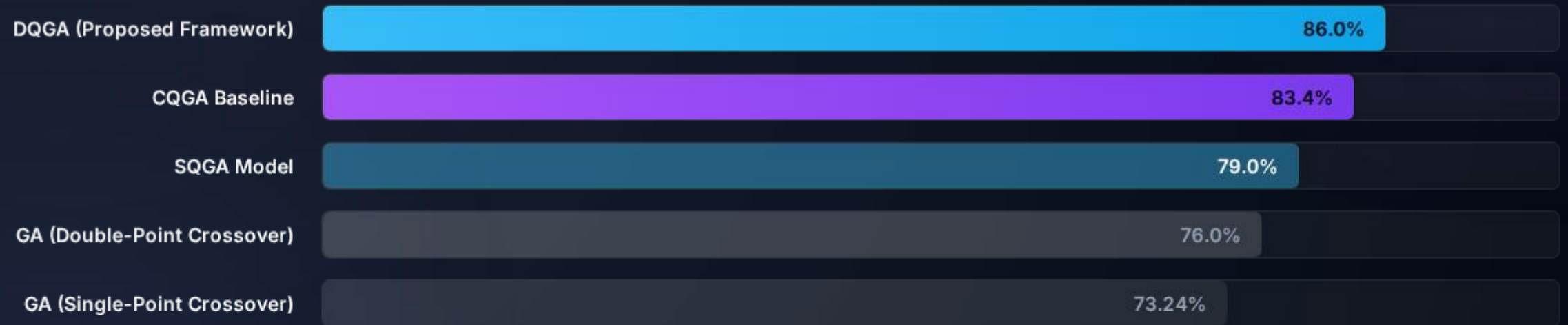
- If the current makespan is worse, Delta, theta values are scaled up to push the amplitudes out of sub-optimal local basins.
- If the schedule meets requirements, the step size is tightened to focus on fine-tuning details.

Experimental Setup & Configurations

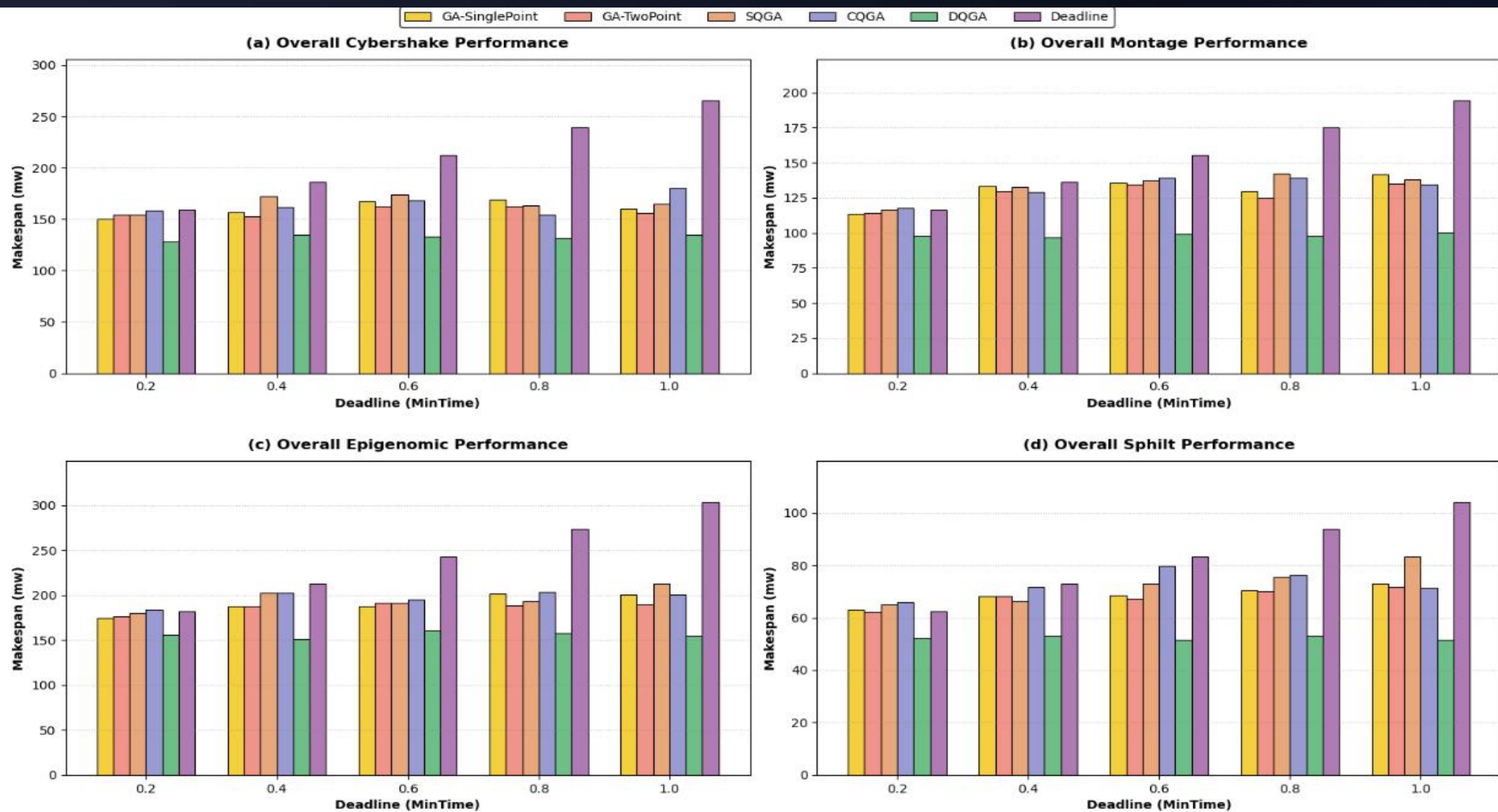
Computing Tier	Instance Name	Cost (USD/Hr)	CPU Cores	Memory (GB)	Bandwidth (Mbps)
Fog Tier (Low Latency)	t3.medium	\$0.0416	2	8.0	500
Fog Tier (Balanced)	t3.large	\$0.0832	2	8.0	1000
Fog Tier (High Performance)	t3.xlarge	\$0.1664	4	16.0	1500
Cloud Tier (Standard)	c4.large	\$0.1000	2	3.75	500
Cloud Tier (Extreme)	c4.xlarge	\$0.1990	4	7.50	800

* Pricing and instances modeled strictly on standard Amazon Web Services (AWS) EC2 instances as detailed in the report.

Planning Success Rate (PSR) Comparison



DQGA achieves an outstanding planning success rate (PSR) of 86.0%, consistently outperforming all traditional and baseline quantum scheduling models across diverse network workloads.



Conclusion and Future Scope

- **Conclusion:** The **DQGA** framework successfully stabilizes quantum collapse through Algorithm 1.
 - The results show that in most cases, DQGA outperforms existing algorithms. DQGA achieves an average of 86% of PSR, where as CQGA has 83.4%, SQGA has 79% and GA with Double point crossover has 76% and GA with Single point crossover has 73.24% PSR.
- **Future Scope: Multi-Cloud Expansion:** additional investigations will be conducted to address the multi-objective workflow scheduling problem in a cloud-fog environment

Thank You!